

063-0605-23L

Computational Structural Design I

Computational 2D Graphic Statics

Dr. Lluís Enrique Monzó

Block Research Group (Prof. Dr. Philippe Block)

HS2023



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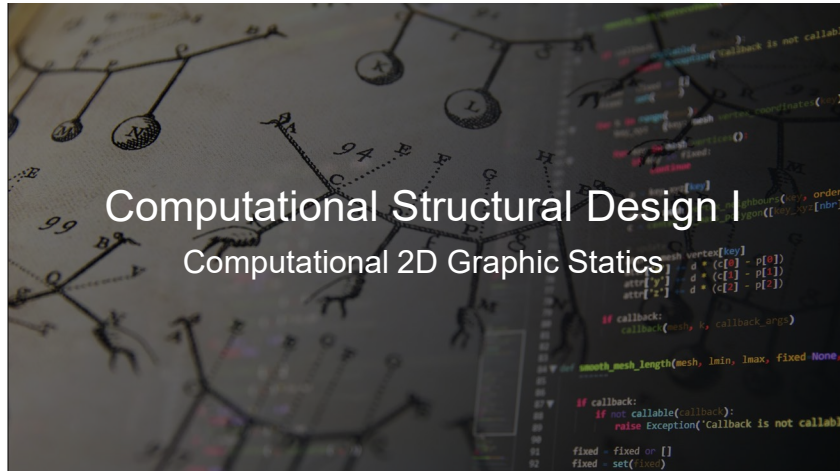
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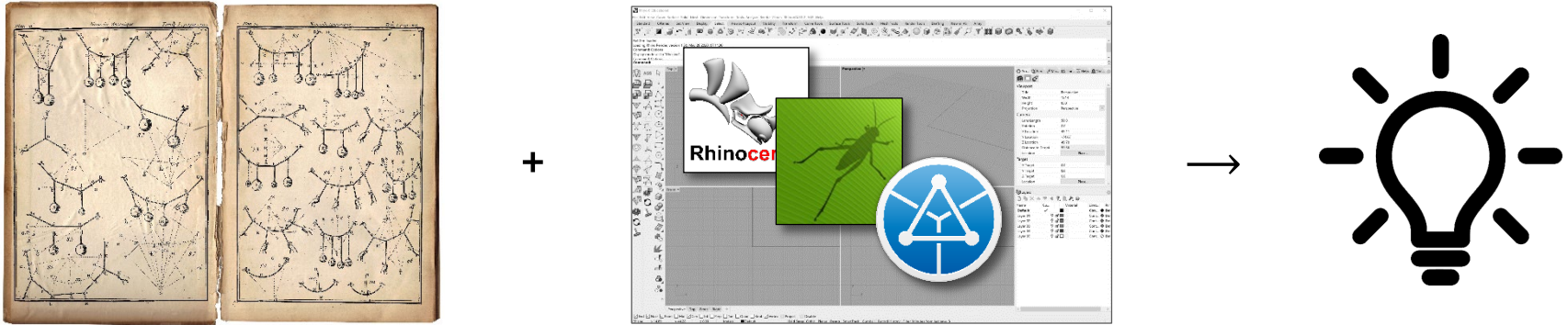




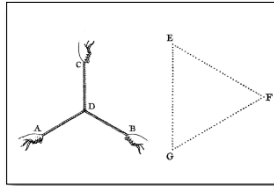
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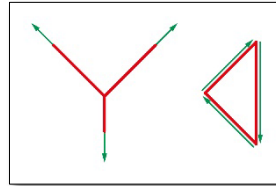
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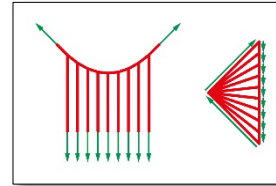
Explore new **structural design opportunities**
by combining **graphic statics** with **computational tools**



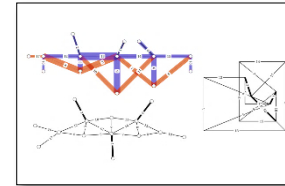
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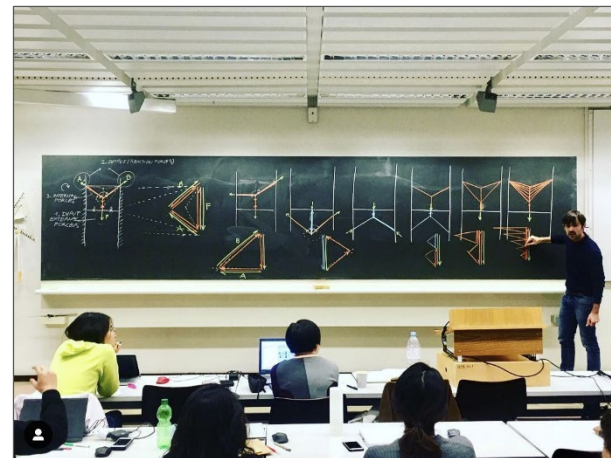
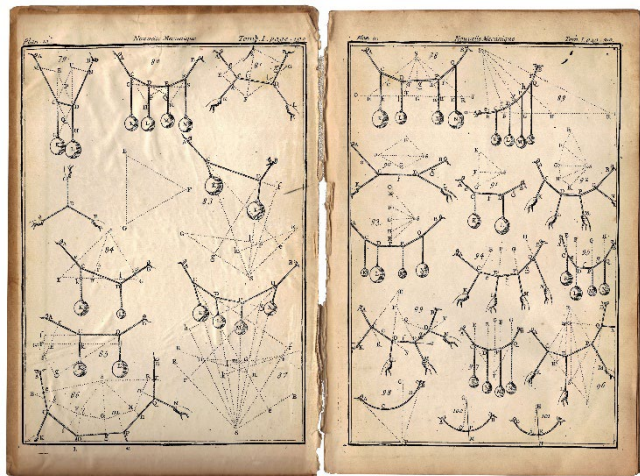
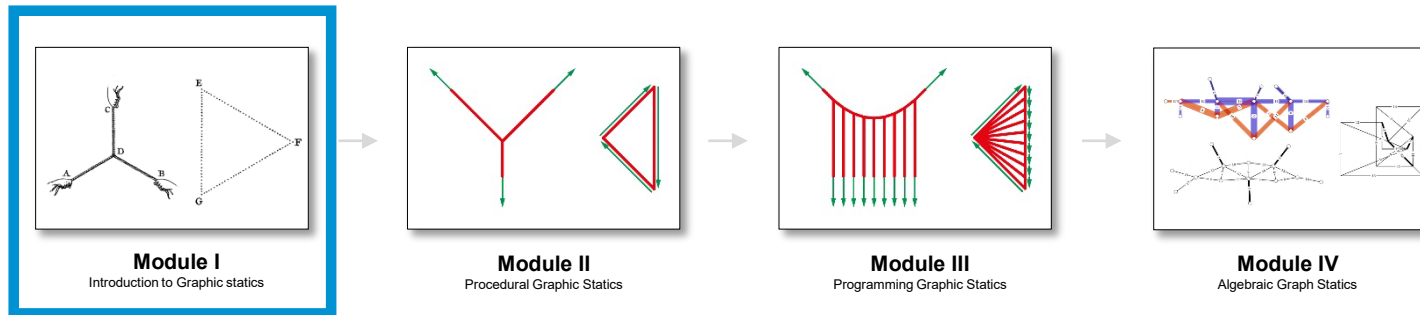
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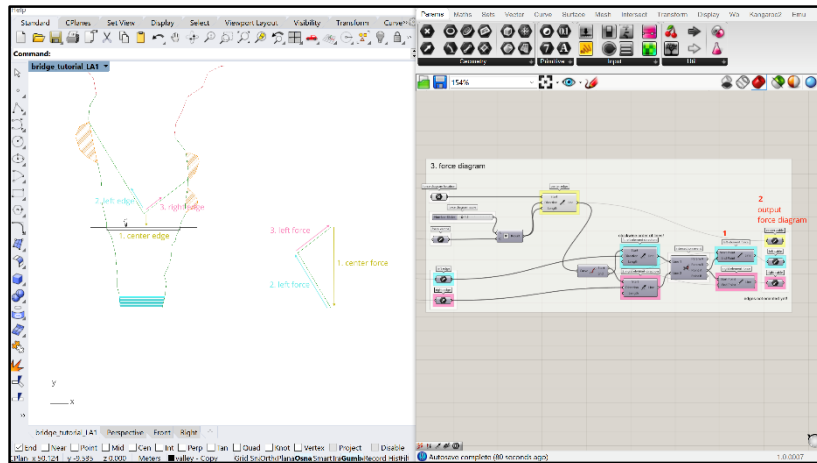
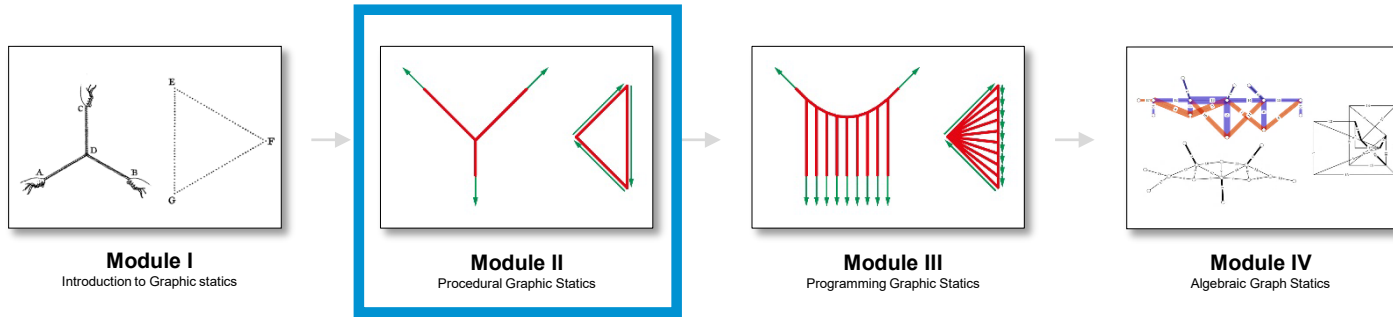


Module III
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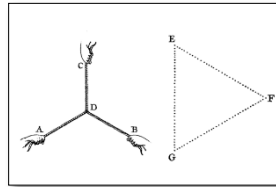
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Algebraic Graph Statics



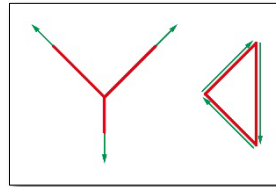


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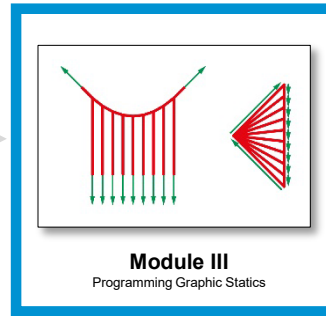




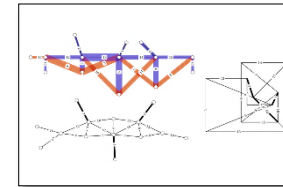
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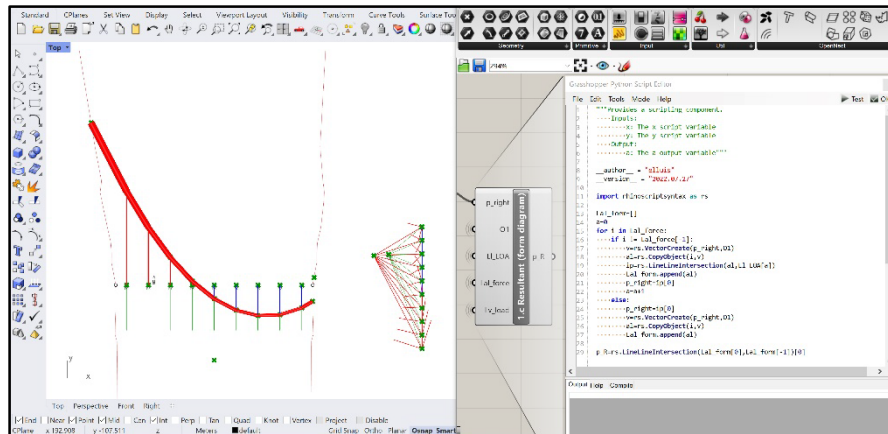
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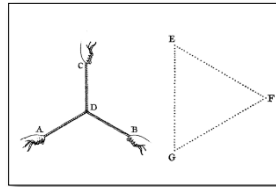


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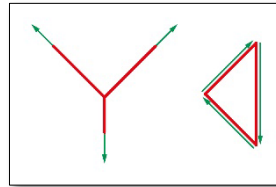


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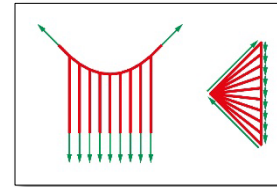




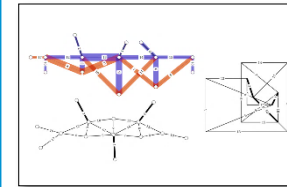
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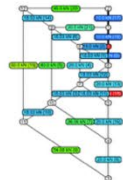
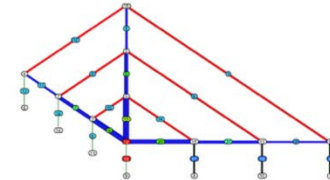
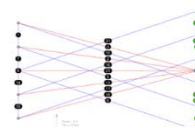
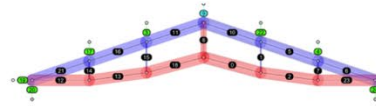
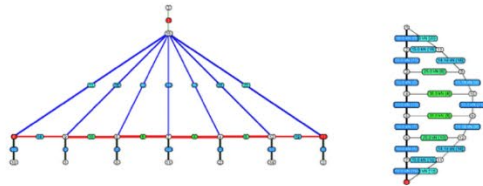
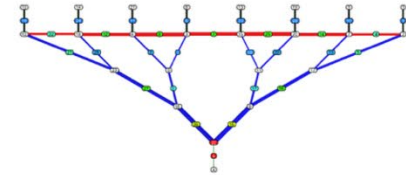
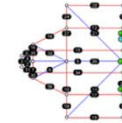
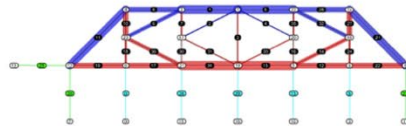
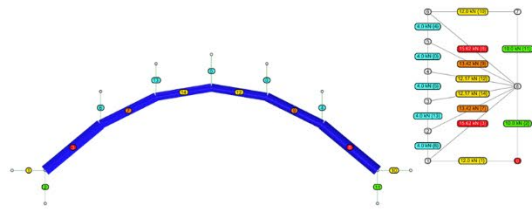
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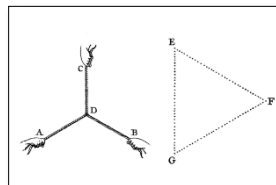


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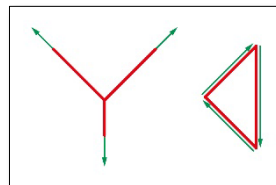


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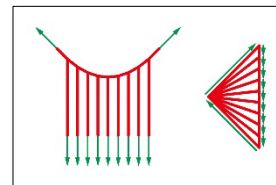


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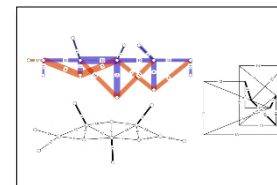
Introduction to Graphic statics

**Module II**

Procedural Graphic Statics

**Module III**

Programming Graphic Statics

**Module IV**

Algebraic Graph Statics



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By the end of the course, students should be able to do the following:

- Use graphic statics for the form finding and analysis of structures.
- Understand the goal and structure of an algorithm.
- Create algorithms based on graphic statics to explore equilibrium systems.
- Explain the potential of studying graphic statics models using computational tools.
- Use Interactive Graphic Statics (IGS2) for the analysis, form finding and design of 2D structures.
- Apply computational graphic statics in design contexts.
- Basic use of Rhinoceros and Grasshopper.
- Demonstrate elementary skills in Python scripting.

Module	Week		Topic
I	Week 1 (22/9)	Lecture	Introduction to graphic statics
		Tutorial	Fundamentals of graphic statics Introduction to Rhino and Grasshopper
		Intro exercise	EX I-1: Graphic statics in Rhino EX I-2: Grasshopper basics
II	Week 2 (29/9)		Quick recap of the last exercise
		Lecture	Algorithmic design & thinking
	Tutorial	Single node bridge in Grasshopper	
	Week 3 (06/10)	Intro exercise	EX II-1 : Single node bridge in Grasshopper
		Work session	
	Week 4 (13/10)		Quick recap of the last exercise
		Tutorial	Form-finding of funicular structures using graphic statics Form-finding of funicular structures using graphic statics in Grasshopper
	Week 5 (20/10)	Intro exercise	EX II-2 : Single node bridge in Grasshopper
		Work session	
Week 6: seminar week			
III	Week 7 (03/11)		Quick recap of the last exercise
		Lecture	Computational graphic statics
		Tutorial	Programming in Grasshopper Python
	Week 8 (10/11)	Intro exercise	EX III-1: Programming in Grasshopper Python
			Quick recap of the last exercise
		Tutorial	Multi-node bridge in Grasshopper Python
		Intro exercise	EX III-2: Multi-node bridge in Grasshopper Python
		Work session	
	Week 9 (17/11)		Quick recap of the last exercise
Intro exercise		EX III-3: Multi-node bridge in Grasshopper Python	
IV	Week 10 (24/11)		Quick recap of the last exercise
		Lecture	Algebraic Graph Statics (AGS)
		Tutorial	Analysis of trusses using graphic statics Analysis of trusses using Interactive Graphic Statics (IGS2)
		Intro exercise	EX IV-1: Analysis of trusses using Interactive Graphic Statics (IGS2)
	Week 11 (01/12)	Work session	
			Quick recap of the last exercise
		Tutorial	Form-finding using Interactive Graphic Statics (IGS2)
		Intro exercise	EX IV-2: Form-finding using Interactive Graphic Statics (IGS2)
	Week 12 (08/12)	Work session	
			Quick recap of the last exercise
		Intro exercise	EX IV-3: Design exploration with Interactive Graphic Statics (IGS2)
		Work session	

063-0605-23L (CSDI) - 2023
Computational 2D Graphic Statics

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COURSE MATERIAL

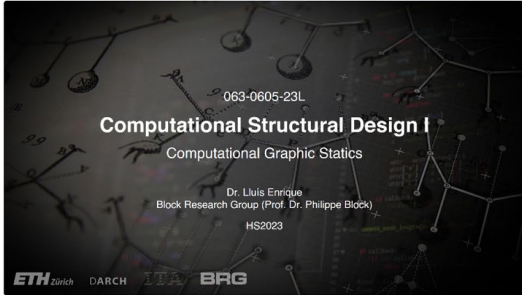
I. Introduction >

APPENDIX

References

Slack

About



Welcome to the course Computational Structural Design I: Computational 2D Graphic Statics!

This course presents new structural design opportunities that emerge when graphic statics, an intuitive equilibrium-based method for the form-finding, analysis and design of structures, is combined with computational tools. After a review of graphic statics fundamentals and an introduction to basic parametric tools, the course focuses on studying different computational approaches to program graphic statics algorithms for the form-finding and analysis of structures, from linear (procedural) algorithms to programs with more complex data structures. More specifically, the students will create graphic statics interactive drawings using Grasshopper, read and code basic graphic statics algorithms using Python, and explore the design space offered by Interactive Graphic Statics (IGS2), a tool developed by the Block Research Group for the analysis, form-finding and design of 2D equilibrium structural systems. The practical potential and relevance of these methods will be demonstrated through various design-oriented tutorials and exercises.

General information

- **Name:** 063-0605-23L : Computational Structural Design I
- **Semester:** Autumn Semester 2023
- **Lecturer:** Dr. Lluis Enrique Monzó
- **Date/time:** Fridays, 09:45 - 12:00
- **Location:** HPT C 103

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ON THIS PAGE

Welcome to the course Computations...

General information

The screenshot shows the Slack interface for a user profile named 'enrique'. The left sidebar is dark purple and contains navigation options: 'CSDI' (with a dropdown arrow), 'Upgrade Plan', 'Threads', 'Direct messages', 'Mentions & reactions', 'Drafts & sent' (with a '2' badge), 'Canvases', 'Slack Connect', 'Files', and 'More'. Below these are 'Channels' (including '# general', '# instructors', and 'Add channels') and 'Direct messages' (including 'Slackbot' and a list of contacts: 'enrique you' (highlighted), 'chaoyudu', 'Li Chen', 'Michelle Kraus', 'Ming Yang Wang', 'Philipp Eitel', 'selina', and 'Theo Mayer'). The main area is white and shows the profile for 'enrique'. At the top, there's a search bar with 'Search CSDI' and a '+ Add a bookmark' button. The profile section includes a blue profile picture placeholder, the name 'enrique (you)' with a green online status dot, and a bio: 'This is your space. Draft messages, list your to-dos, or keep links and files handy. You can also talk to yourself here, but please bear in mind you'll have to supply both sides of the conversation.' Below the bio are two buttons: 'Edit Profile' and 'Upload Profile Photo'. At the bottom is a text input field with the placeholder 'Jot something down' and a rich text toolbar with icons for adding attachments, text formatting, emojis, mentions, links, and code blocks.

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Week 1 Friday, September 22nd

9:45 – 10:30 Welcome & introduction
Lecture: Introduction to graphic statics

10:30 – 10:45 Break

10:45 – 11:30 **Tutorial: Fundamentals of graphic statics**
Tutorial: Introduction to Rhinoceros

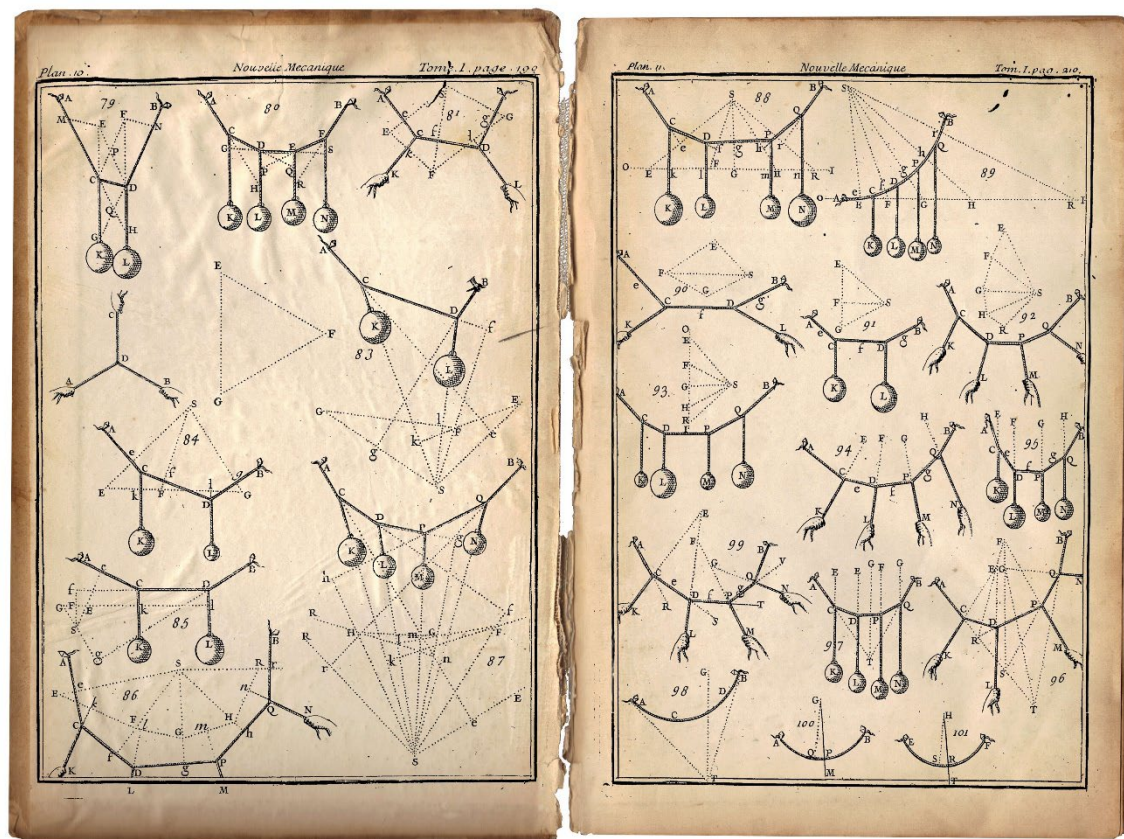
11:30 – 11:45 Break

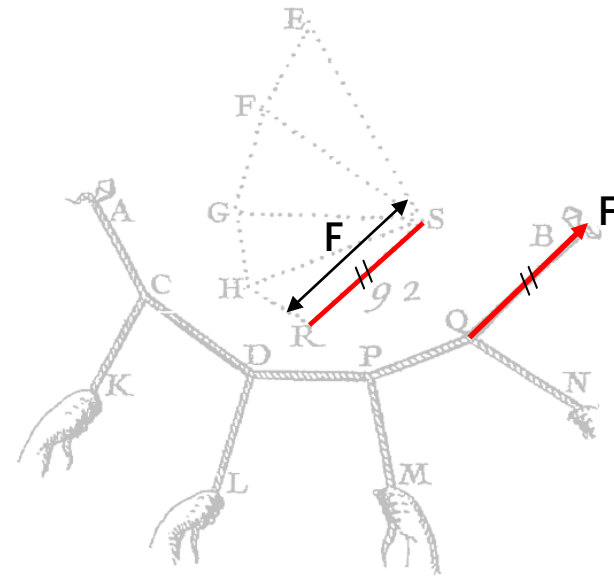
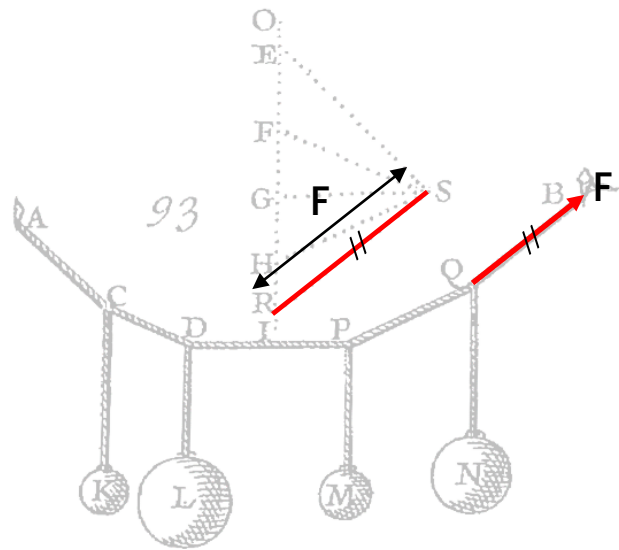
11:45 – 12:30 **Tutorial: Introduction to Grasshopper**
Introduction exercises

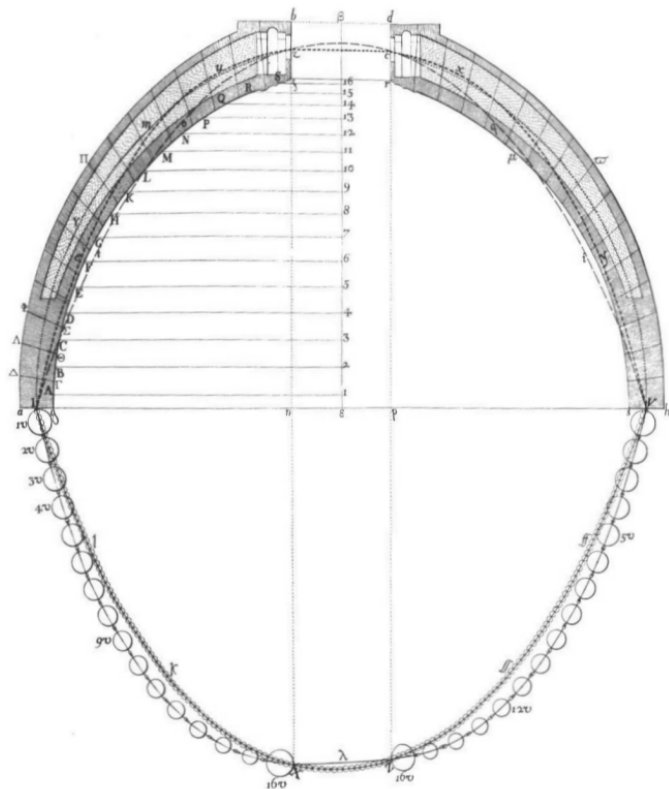
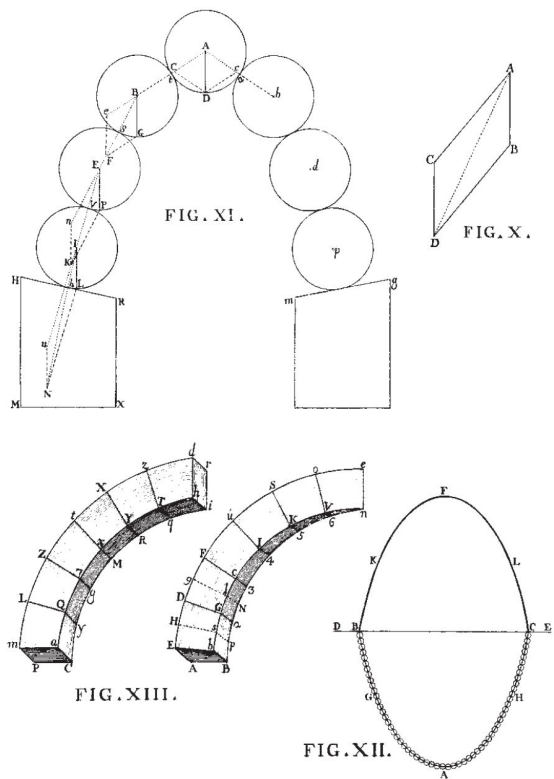
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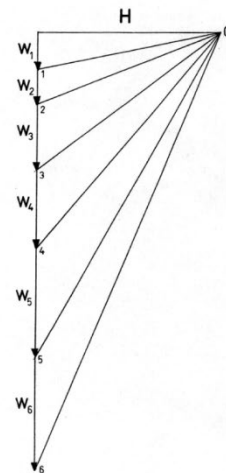
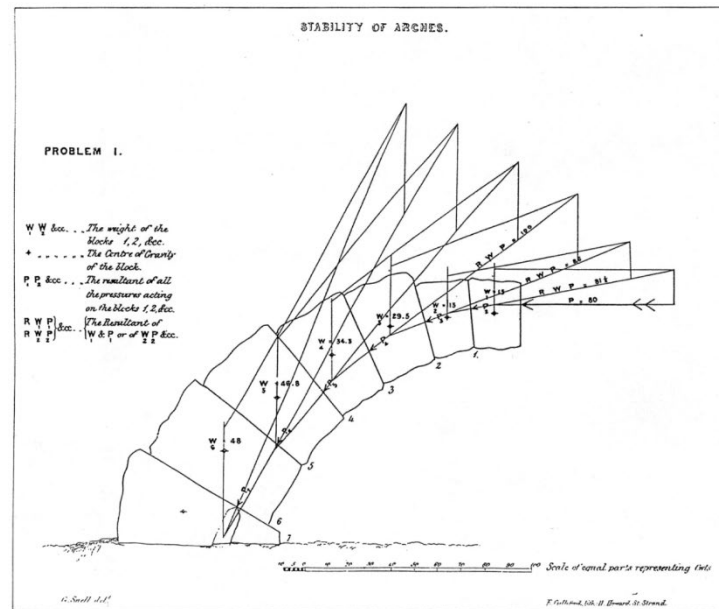
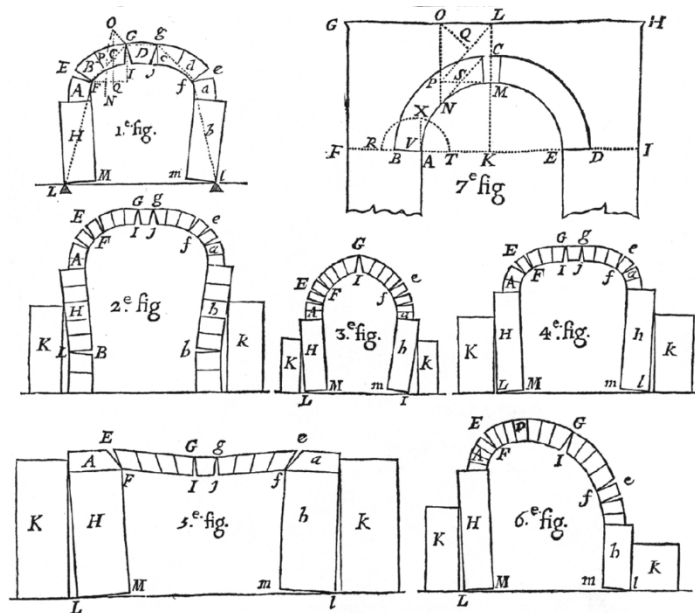
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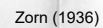
Dr. Lluís Enrique Monzó

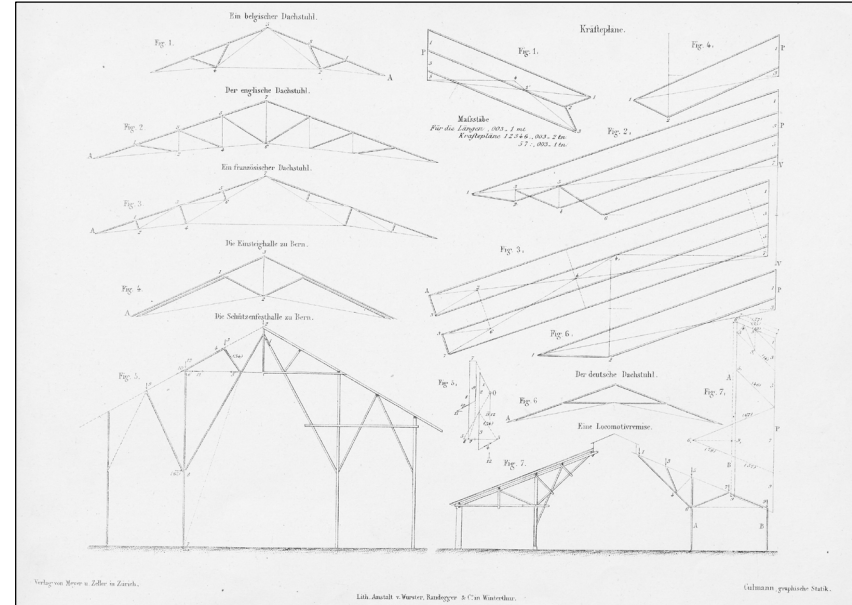
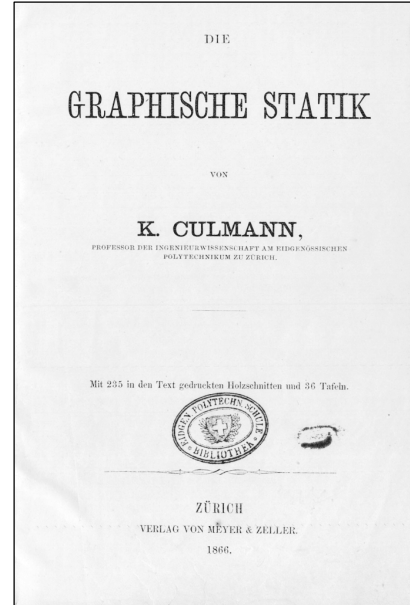












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the upper extremity of the segment 1 meets the upper extremity of the segment 2; and the straight line 18 through the point (16, 1), which is both the lower extremity of the segment 16 and that of the segment 1.



Fig. 8a.

Pass on to the joint (2, 17, 19, 30). Draw 19 through the point (17, 18), and 20 through the point (2, 3), the lower end of 2 and upper extremity of 3; and we obtain the polygon 2, 17, 19, 20, which is a rectangle.

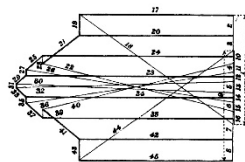


Fig. 8b.

Construct the polygon corresponding to the joint (16, 18, 19, 21, 22). For this purpose draw 21 through the point (19, 20), and 22 through the point (15, 16); we thus obtain a crossed pentagon. Continue to deal in the same manner with each of the points of application of the forces 3, 15, 4, 14, 13, 6, 12, 6, 11, 7, 10, 9, taken in succession.

Since the diagram *a*, which represents the skeleton of the structure and all the external forces, has for its axis of symmetry the vertical which passes through the centre of the figure, the diagram *b* has for its axis of symmetry the median horizontal line. For example, the triangle 9, 45, 44 is sym-

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metrical to the triangle 1, 17, 18; the rectangle 8, 45, 43, 42 to the rectangle 2, 17, 19, 20; and so on.

All the upper bars are in compression, and all the lower ones are in tension.

The diagonals and contra-diagonals are all in compression; finally two of the verticals 23, 39 are in tension, and all the rest in compression.

44. Figure 9a* represents one half of a locomotive shed. The external forces are the weights 1, 2, 3, 4, 5 applied at

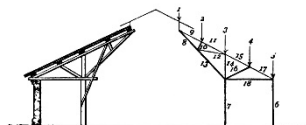


Fig. 9a.

the upper joints of the frame, and the reactions 6 and 7 of the wall and column. Again, all the external forces are parallel, and consequently the polygon of forces reduces, in diagram *b*, to one straight line. The force 8 (taken in the opposite sense to that in which it really acts) is equal to a certain part of the weight 5; by adding the difference to the other weights we get the magnitude of the force 7.

In the diagram *b* the direction of the lines 8 and 13 coincide; the first is a part of the second. Here then we have for the polygon corresponding to the joint (8, 10, 12, 13) one of those degenerate forms about which we spoke in Art. 33; the polygon is in fact a quadrilateral 8, 10, 12, 13, having three of its vertices (13, 9), (9, 10), (12, 13) in one straight line.

The polygon 5, 17, 18, 6, corresponding to the point where

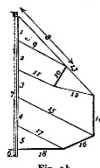


Fig. 9b.

* This example is taken from PLATE of the *Atlas of Graphical Statics* of CREMONA, 1st edition. As previously stated, the two diagrams are not rigorously reciprocal.

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the roof is supported by the wall, presents an analogous degenerate form, since the vertices (6, 5), (upper point of the segment (6), (5, 17), and (18, 6) all lie in the same straight line.

The lower bars 8, 13, 18 are in compression, as well as the diagonals 10, 14, 16, the column 7 and the wall 6; while the upper pieces 9, 11, 15, 17 and the diagonal 12 are in tension.

45. Diagram *a* of Fig. 10 represents a truss at the upper joints of which are applied the oblique forces 1, 2, ..., 7, which

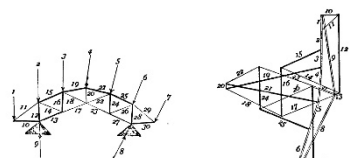


Fig. 10a.

may be considered as the resultants of the dead-loads and wind pressure; the forces 8, 9 represent the reactions at the supports.

The polygon of external forces is drawn in diagram *b* with double lines.

We construct successively the triangle 1, 10, 11, the quadrilateral 9, 10, 12, 13, the pentagon 2, 11, 12, 14, 15, the quadrilateral 13, 14, 16, 17, the crossed pentagon 3, 15, 16, 18, 19; the crossed quadrilateral 4, 19, 20, 21, the pentagon 17, 18, 20, 22, 23, and so on.

The upper bars 16, 19, 21, 23 are in compression, as well as the lower bars 10, 13, 18, and the verticals 12, 16, 24, 28; whilst all the remaining bars of the structure are in tension.

46. The diagram *a* of Fig. 11 represents a suspension bridge, loaded at each of its upper joints with weights 1, 2, ..., 8, and at each of its lower joints with weights 10, 11, 21, ..., 16; the weights are kept in equilibrium by the two

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oblique reactions 9, 17 at the two extreme points of the structure*.

The polygon of external forces has its first eight sides in succession along the same vertical straight line, and its seven

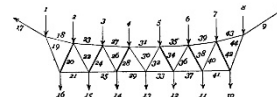


Fig. 11a.

last sides situated in another vertical straight line. The oblique sides 9 and 17 intersect, so that the polygon is a crossed one. We construct successively the polygons 1, 17, 19, 18; 16, 19, 20, 21; 2, 18, 20, 22, 23; 15, 21, 22, 24, 25; 3, 23, 24, 26, 27; and so on; most of which are crossed.

Diagram *b* shows that the upper bars are all in tension, and that the tension decreases from the ends towards the

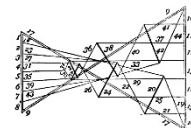


Fig. 11b.

middle of the structure; the bars of the lower boom are also all in tension, but in them the tension decreases from the middle towards the ends.

The extreme diagonals and contra-diagonals are in tension; in the portion situated to the left of the axis of symmetry, the diagonals or braces are alternately in tension and compression; similarly they are on the right but in the reverse order. Considering separately the ties and struts, we see that the internal

* This example is analogous to one of those studied by Maxwell in his memoir of 1870.

PLATE VII.

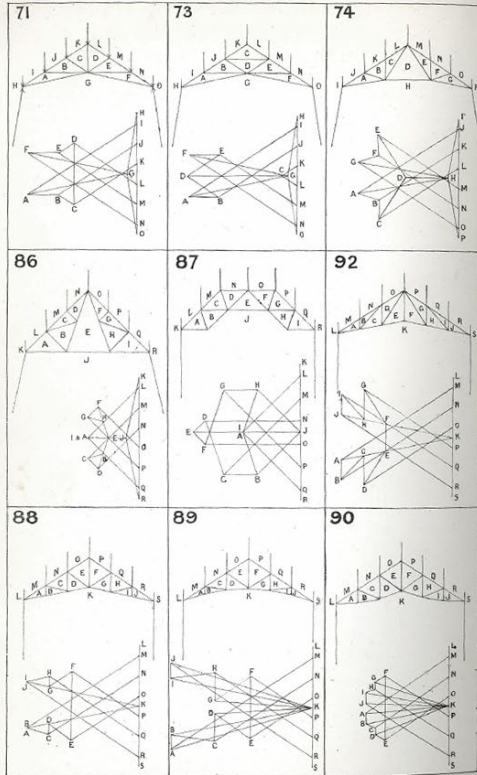
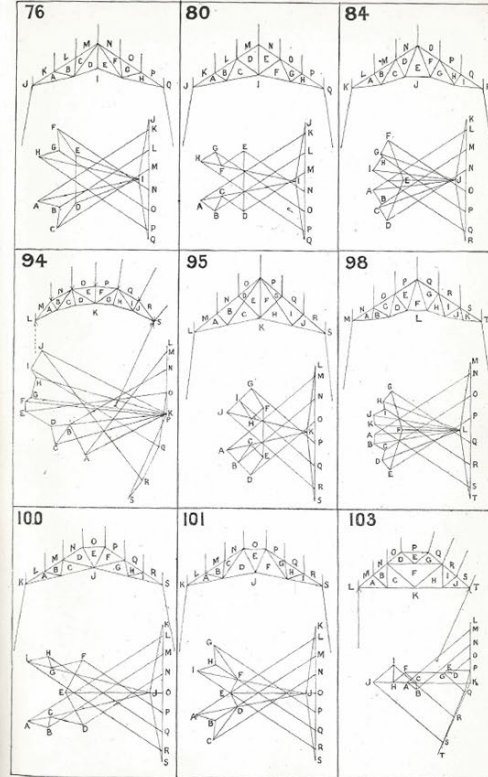
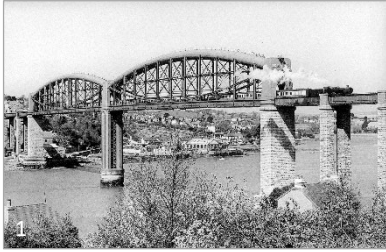


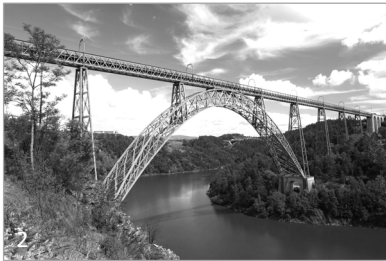
PLATE VIII.





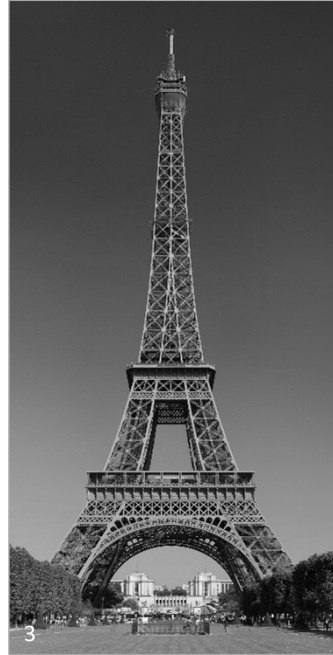
Royal Albert Bridge (1859)

Isambard Kingdom Brunel



Garabit Viaduct (1884)

Maurice Koechlin



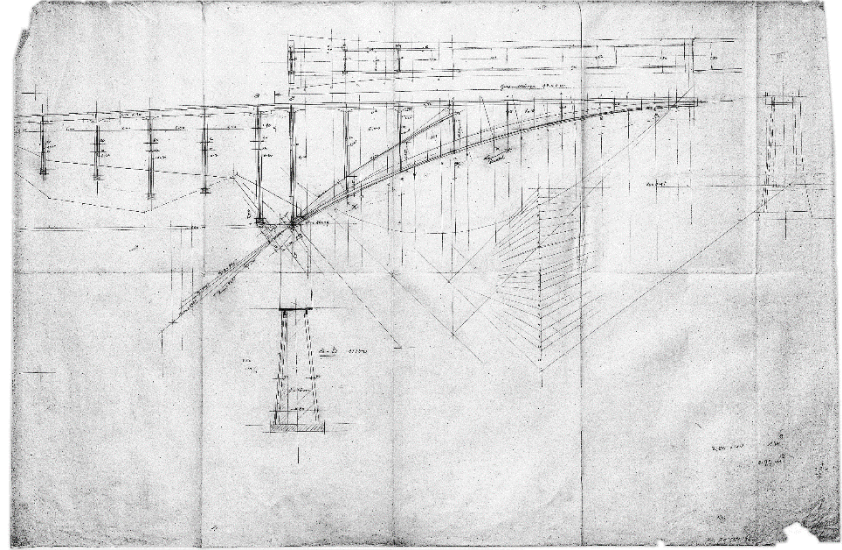
Eiffel Tower (1887)

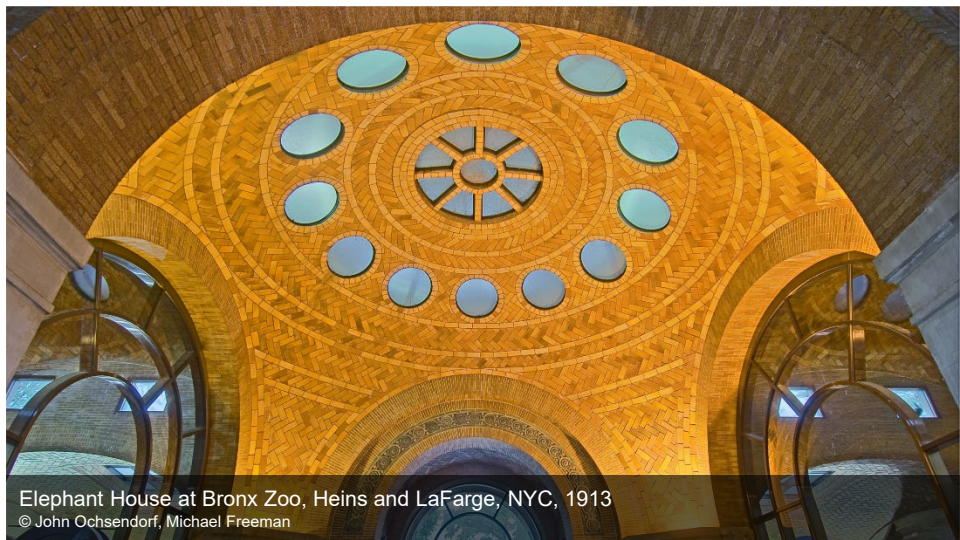
Maurice Koechlin, Émile Nouguier

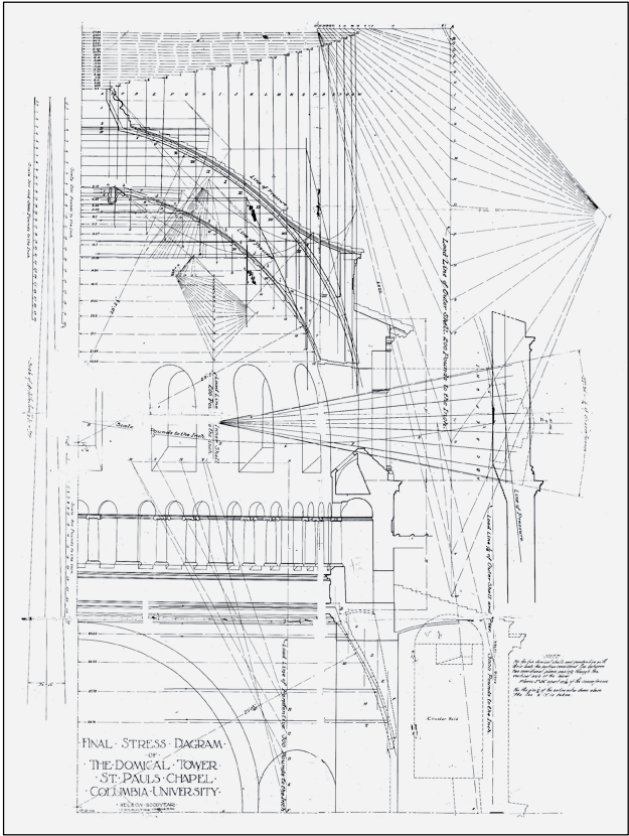


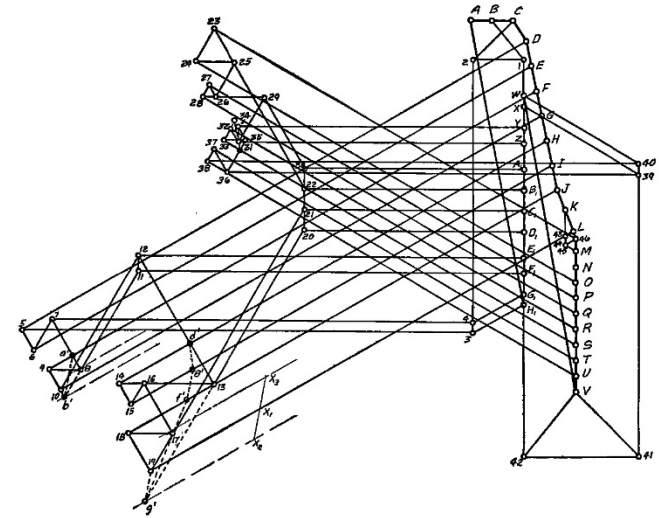
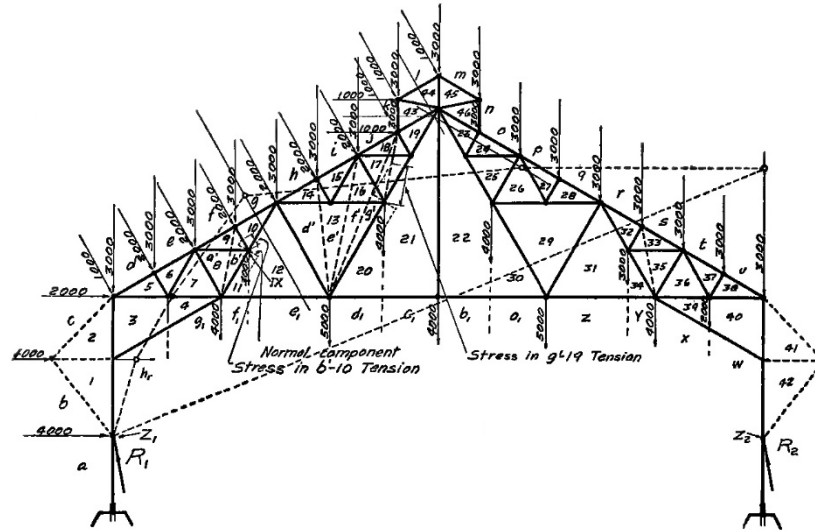
Forth Bridge (1890)

Sir John Fowler, Sir Benjamin Baker









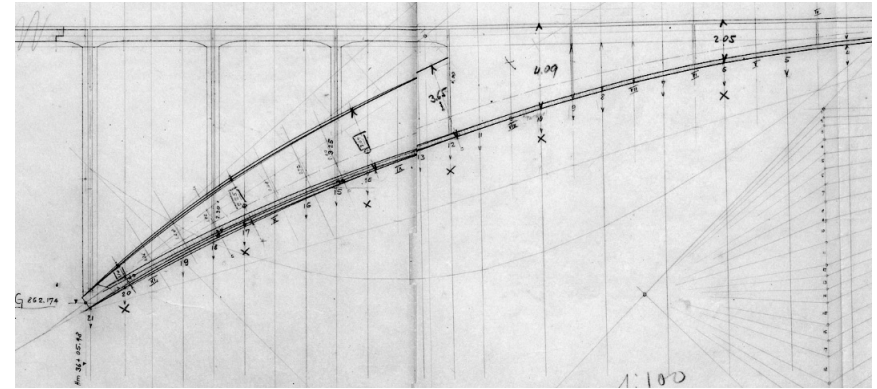
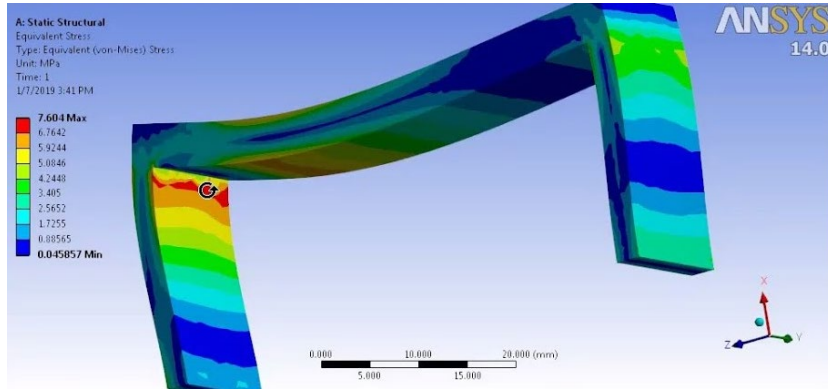
TOTAL LOAD ON
ONE HANGER

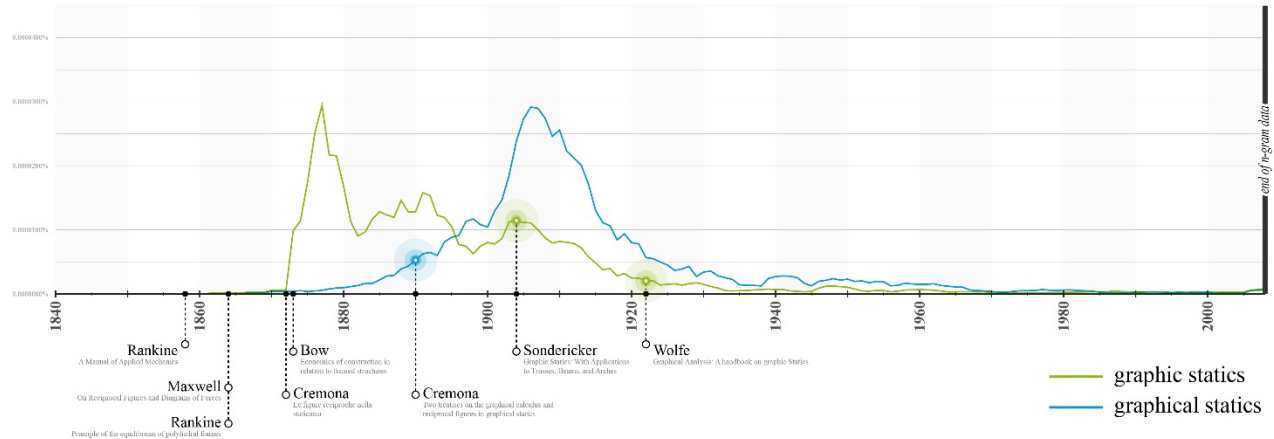
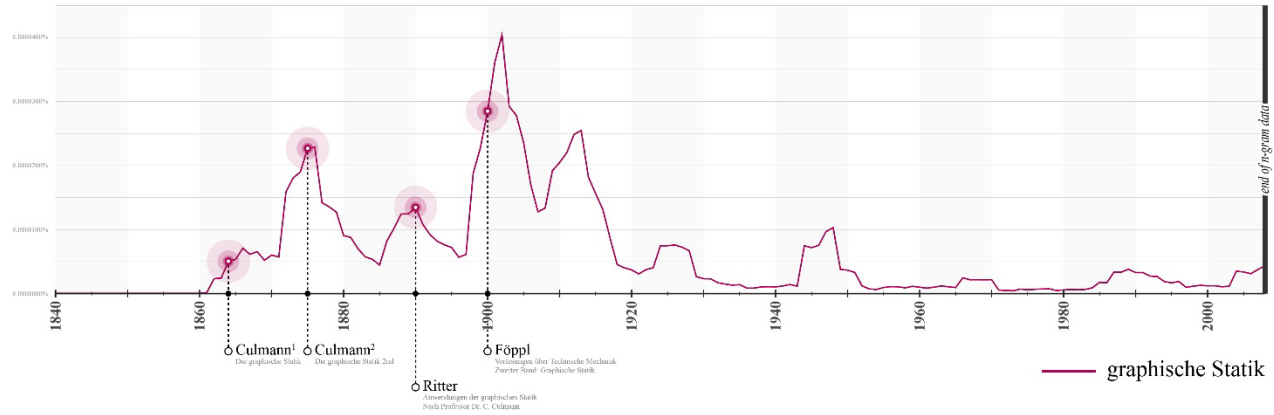
LOAD = 60 psf

$P = 60 \text{ lb} (400 \text{ ft})$

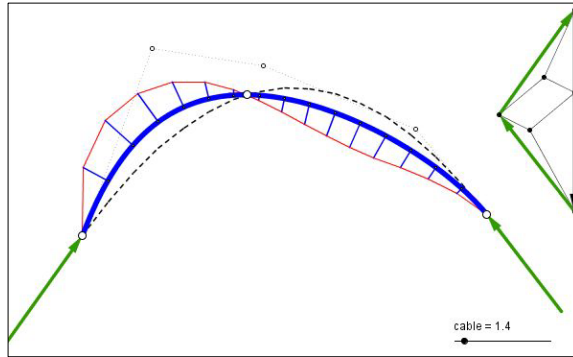
$P = 24,000 \text{ lb}$
 $= \underline{\underline{24 \text{ K}}}$



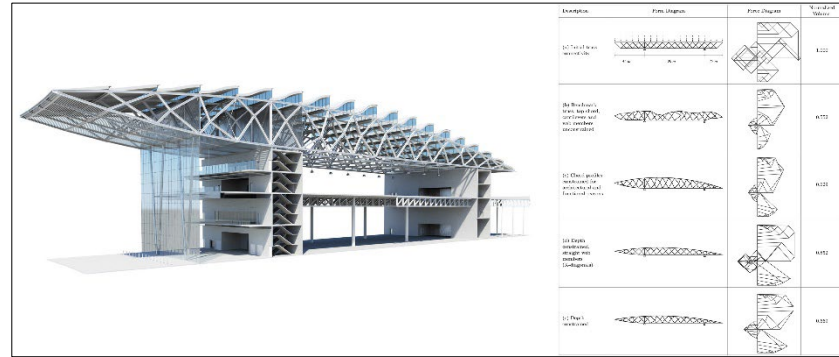




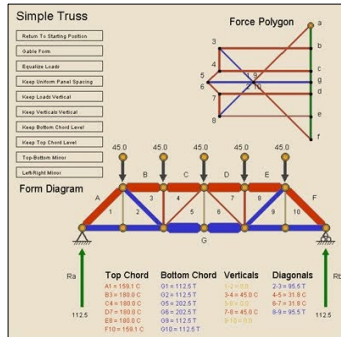


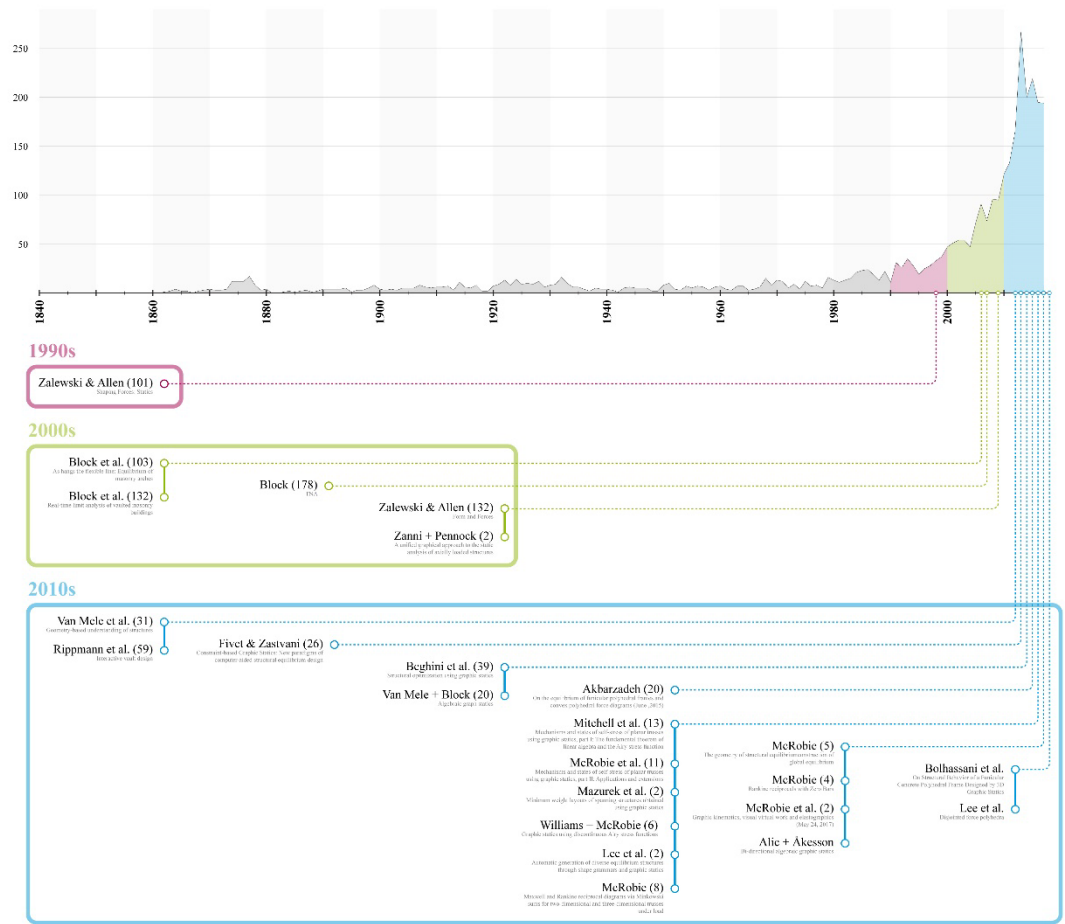


eEQUILIBRIUM
Van Mele et al. (2009-)

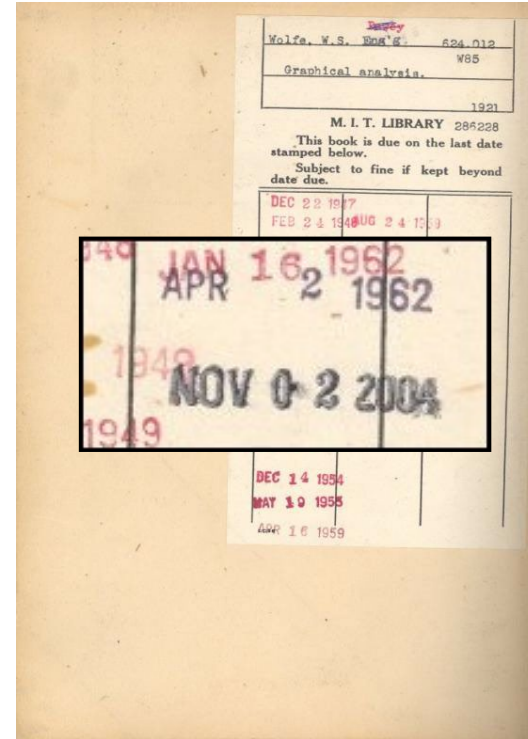
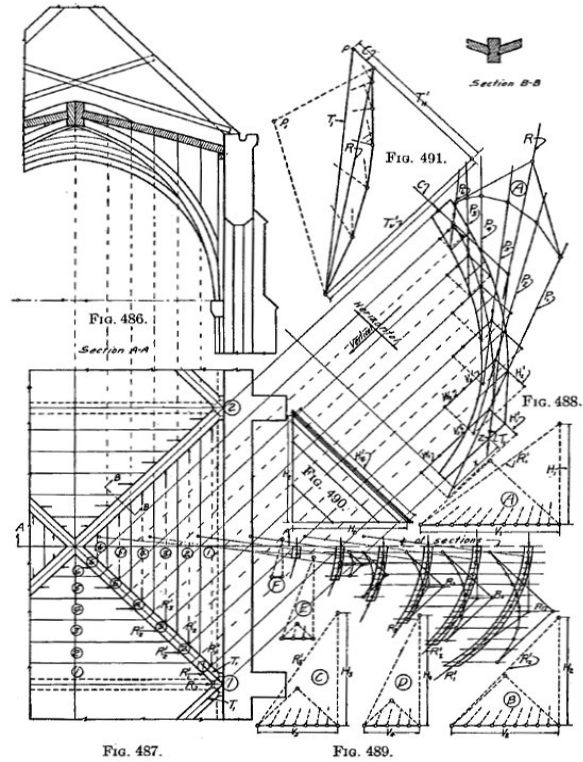


Structural Optimization using Graphic Statics
Beghini, et al. & SOM (2013)





Timeline of graphic statics publications (early 2018) | from PhD dissertation by Lee (2018)











Karl Culmann

- Founder of graphic statics: “Die graphische Statik” (1864-1866)
- First Professor of Civil Engineering at the ETH Zurich (1855-1883)
- Director of the Polytechnic (1872-1875)

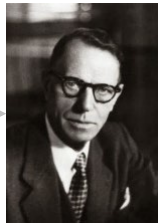


Karl Wilhelm Ritter (ETHZ diploma, 1868)

- Professor of graphic statics and bridge construction (appointed in 1882)
- Director of the Polytechnic (1887-1891)
- Extended applications of Culmann’s graphic statics:
“Anwendungen der graphischen Statik nach Professor Dr. Culmann” (1888-1906)



Robert Maillart
(ETH diploma, 1894)



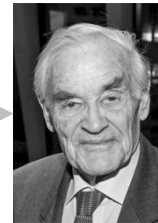
Othmar Ammann
(ETH diploma, 1902)



Pierre Lardy
(ETH diploma, 1928)



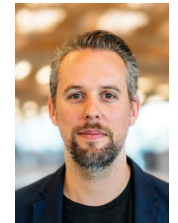
Heinz Isler
(ETH diploma, 1950)



Christian Menn
(ETH diploma, 1950)



Joseph Schwartz
(ETH professor, 2008-)



Philippe Block
(ETH professor, 2009-)

